

An Adaptive RRT Based on Dynamic Step for UAVs Route Planning

Na Lin

*School of compute science,
shenyang aerospace university
Shenyang, liaoning province. 110136, china
Lin_na2000@163.com*

Ya-lun Zhang

*School of compute science,
shenyang aerospace university
Shenyang, liaoning province. 110136, china
Zhangyalun0710@qq.com*

Abstract—Executing task in the complex environment, the traditional RRT algorithm is easy to make the tree fall into local minimum area which resulted in poor efficiency. In order to solve the problem, this paper proposes an adaptive RRT algorithm based on dynamic step for path planning and uses the cubic polynomial BSpline curve fitting to smooth the planned path. The simulation results show that the number of explore failures reduced 35186 times, consuming time reduced 963 ms, distance reduced 33 km at the 3000th iteration by using the adaptive RRT based on dynamic step comparing to the traditional RRT algorithm.

Keywords— Unmanned aerial vehicle (UAV); dynamic step; rapidly-exploring random tree(RRT); BSpline curve; route planning

I. INTRODUCTION

Owing to its small size, low cost, convenient use, low requirement of combat environment, strong survivability in battlefield, unmanned aerial vehicle (UAV) plays an irreplaceable role in some highly dangerous missions^[1]. However, many traditional algorithms cannot meet the actual requirement for UAVs route planning in the complex environment because of their own disadvantages and shortages.

For example, the A* algorithm^[2] uses the strategy of nodal expansion to route planning, but there exists the problem of combinational explosion when excuting task in complex environment due to the lack of nodal expansion resulting in the planning time drastically expands and low efficiency. Due to the effect of unknown factors in the complex environment, the genetic algorithm^[3] easily falls into the blind searching condition in the evolutionary process, which leads to low searching efficiency. The ant colony algorithm^[4] uses pheromone as the basis for the choice of path, easily falls into local optimum area, resulting in the relatively long searching time and low programming efficiency. So, it's not suitable to conduct planning in the complex environment. RRT algorithm^[5] doesn't need to construct the tactical environment, only needs to get a path through the selection of random exploratory node, so it is convenient to realize. However, the RRT algorithm is easy to fall into local minimum area if UAV doesn't not know the distribution of obstacles in the complex environment, although it can jump out of local minimum area

after many selections of random node, it will reduce the planning efficiency.

In recent years, in order to make up for the shortcomings of traditional RRT algorithm, more and more improved RRT algorithms are used to route planning. For example, in the route planning, the two-way RRT and RRT-Connect^[6-7] (CRRT) begin to construct the tree from the starting point to the target point at the same time until these two trees have intersection. Thus, we can find a feasible path from the starting point to the target point. These improved RRT algorithms can improve the efficiency of path planning, reduce the required planning time, but there often exists some gaps between these planned paths and the optimal path, and their cost is relatively high. The RRT*, extend RRT (ERRT) and Closed-loop RRT (CL-RRT)^[8-11] algorithms are proposed in order to make the planned path much closer to the optimal path and improve the stability of path planning. However, they are not conducive to solve the situation of dynamic threat because of the increased planning time resulted from the increased number of failed searching in the complex environment.

Summarizing these above problems, this paper presents an adaptive RRT algorithm based on dynamic step (DRRT) for path planning. The improved RRT algorithm can use the feedback information of tree for the environment to adjust exploratory step and node weight dynamically when UAV carries out a task in the complex environment of multiple obstacles which can reduce the failure number of exploring, improve the speed of exploration and shorten the length of path. In the route planning process, this paper introduces the body constraint condition of UAV^[12] and the cubic polynomial BSpline curve fitting^[13] to smooth the planned path, avoiding the unreasonable of planned path. The simulation results show the effectiveness and feasibility of improved algorithm.

II. THE PATH PLANNING OF UAV

A. Flight environment and task description

According to the known air defense and radar threat, UAV needs to carry out route planning and produce a global optimal or suboptimal path before executing the task. Generally speaking, all kinds of threats can be simplified as a cone or

The Project is supported by Program for Liaoning Excellent Talents in University (LJQ2012011), Natural Science Foundation of Liaoning Province (No. 20102175), Liaoning Province Education Department Research Projects (No. L2010423).

cylinder with different radii. In order to facilitate the simulation, this paper assumes that the UAVs flight height is unchanged, thus, the 3D planning problem can be transformed into plane problem. The threatened species are shown in Table 1:

TABLE I. THE DATA OF THREAT

The type of threat	The radius of threat (km)
Radar	30
Short-range missile	10
Medium-range missile	20

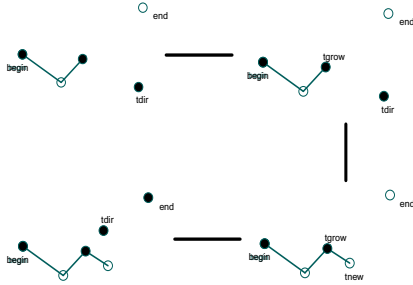
The task of UAVs navigation is flight from the starting point to the target point, and all kinds of threats existing between them such as the threat of radar and missile. The aim of route planning is to search for an optimal path from the starting point to the target point, avoiding all threats at the same time.

B. The traditional RRT algorithm

In recent years, the RRT algorithm has become one of the popular motorial algorithms of planning, especially in the route planning for UAV. The RRT algorithm uses the initial point of the given task as the root node by constructing a growth tree to realize path planning in the battled environment, selecting the growth node of tree and executing the method of collisional detection to get a subset of the battled environment to find an optimal or sub-optimal path.

As shown in figure 1, *begin* and *end* are the starting point and target point of the task respectively, t_{dir} is the random exploratory node, t_{grow} and t_{new} respectively are the growth node and next exploratory node. An expanding process of the tree is shown in the figure below:

FIGURE I. THE ADJUSTMENT OF DYNAMIC STEP



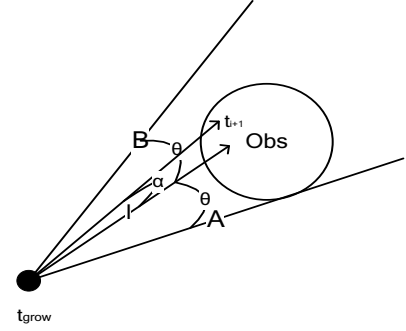
C. The adaptive RRT based on dynamic step

In the figure 2, *Obs* represents the range of obstacle, θ is the maximum turning angle with UAV's constraints. If t_{grow} is the current growth node, then its next exploratory node must fall into the surrounded area by t_{grow} , A and B, d_s is the initial exploratory step, $\vec{l}$ is the current extensional direction of random tree, α is the angle between vector $\overrightarrow{t_{grow}t_{i+1}}$ and $\vec{l}$. If the t_{i+1} is exploratory failed node, the improved RRT dynamically adjust the exploratory step according to the type (1):

$$d = \frac{\alpha}{\theta} \times d_s \quad (1)$$

Otherwise, $d = d_s$.

FIGURE II. THE ADJUSTMENT OF DYNAMIC STEP



Assuming that the current tree named T has M nodes, namely $T = \{t_i | i = 1, 2, \dots, m\}$. The failed searching node is named t_j , $t_j \in T$, f_j is the number of searching failures of t_j . d_{ij} is the distance between aleatoric node t_i and t_j , $d_{ij} = \|t_i - t_j\|$. d_i is the distance between aleatoric node t_i and random exploratory node t_{dir} , $d_i = \|t_i - t_{dir}\|$.

- Initialize the exploratory tree T and step d_s , θ , α , t_l which is the root node of exploratory tree is the starting node of task named t_{start} .

- Compute the random exploratory node t_{dir} :

$$t_{dir} = p \times t_{goal} + (1 - p) \times t_{rand}, (0 < p < 1) \quad (2)$$

(The target point of task t_{goal} , the random point of space t_{rand})

- Compute the exploratory step d :

$$d = \begin{cases} \frac{\alpha}{\theta} \times d_s, & f_j > 1 \\ d_s, & \text{otherwise} \end{cases} \quad (3)$$

- Select the growth node t_{grow} of tree:

$$I_i = \begin{cases} 1/f_j, & d_{ij} < d \\ d_{ij} / (d \times f_j), & d \leq d_{ij} < d \times f_j \\ 1, & \text{otherwise} \end{cases} \quad (4)$$

$$\omega_i = I_i / d_i \quad (5)$$

$$t_{grow} = \arg \max(w_i), t_i \in T \quad (6)$$

- Compute the new exploratory node t_{new} of tree:

$$t_{new} = \begin{cases} t_{dir}, & \|t_{dir} - t_{grow}\| \leq d \\ t_{grow} + (t_{dir} - t_{grow}) / (\|t_{dir} - t_{grow}\|), & \text{otherwise} \end{cases} \quad (7)$$

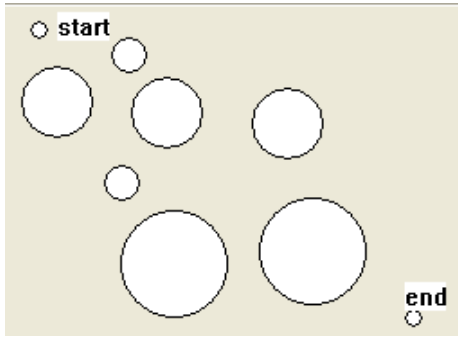
- Updating the growth tree T , determine whether the t_{new} is exploratory failed node, if not, then add the t_{new} to T , and $t_i=NULL$, $f_i=1$. Otherwise, compute the angle α , give up the t_{new} , $t_j=t_{grow}$, f_j+1 , and goto (2).
- Judging whether to reach the target point. If $\|t_{goal} - t_{new}\| < s$, then reach the target point. Otherwise, goto (2). (s is the shortest flight distance of uav)
- Backing from the target point to the root node of tree, and then return the path.

III. SIMULATION

A. The experimental environment

The simulation experiment uses the VC++6.0 platform to realize. The hardware environment: XP SP2, Intel Pentium Dual Core T3400 @ 2.16GHz, 2 GB of RAM, ATI Mobility Radeon HD 2400. In the figure 3, the region of UAVs flight is in the range of 285*185km, there are radar threats, short-range missiles, medium-range missiles, and their radii are respectively 30km, 10km, 20km. There also hides a sudden threat which will randomly appear in the range of the starting point and target point, and its radius is 20km.

FIGURE III. THE MAP OF TASK



B. The roadmap of experiment

This paper respectively uses the traditional RRT, CRRT, adaptive RRT (ARRT), DRRT to simulate the given situation, and all of their iterations are 3000 times.

FIGURE IV. THE SIMULATION RESULT OF RRT

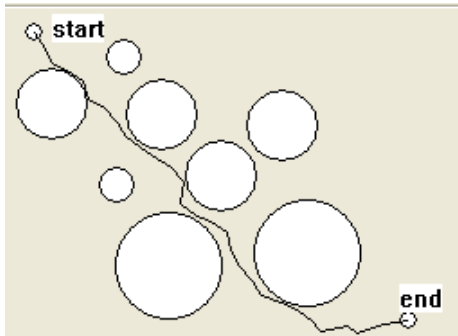


FIGURE V. THE SIMULATION RESULT OF CRRT

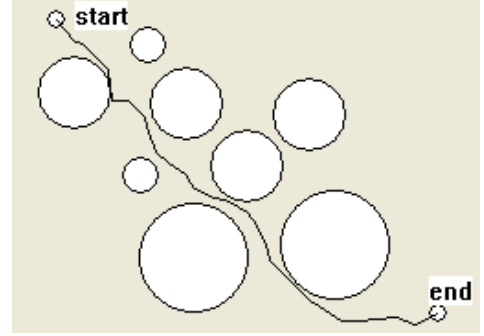


FIGURE VI. THE SIMULATION RESULT OF ARRT

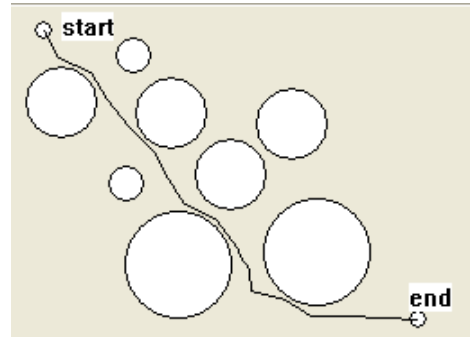
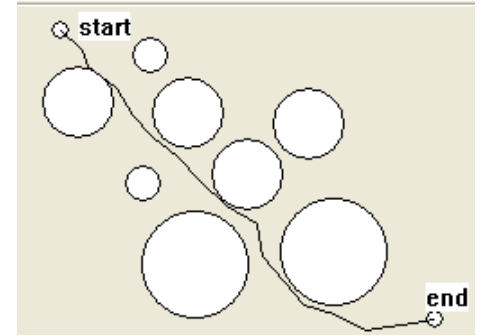


FIGURE VII. THE SIMULATION RESULT OF DRRT



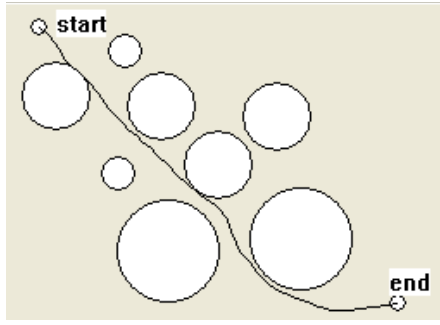
IV. SMOOTHING

From these roadmaps, we can see that all of planned routes exist some sections where angle is too small, so, these routes clearly violate the UAVs body constraints and cannot meet the actual flight requirement. Therefore, this paper uses the cubic polynomial BSpline curve fitting to smooth the planned path.

Assuming that the planned path is formed by $n+1$ points, namely $P_i (i = 0, 1, \dots, n)$, $P_0 = \text{start}$, $P_n = \text{end}$, and each four adjacent nodes can construct a cubic polynomial BSpline curve. The primary function of cubic polynomial BSpline curve is $N_{i,3}(u) = [N_{1,3}(u), N_{2,3}(u), N_{3,3}(u), N_{4,3}(u)]$. So, the BSpline curve can be shown as follow:

$$C_{i,3}(u) = N_{1,3}(u) \times P_{i-1} + N_{2,3}(u) P_i + N_{3,3}(u) P_{i+1} + N_{4,3}(u) P_{i+2} \quad (8)$$

FIGURE VIII. THE SMOOTHED PATH



As we can see in figure 8, the smoothed path eliminates these sections where angle is too small and can meet the actual requirement of UAVs flight.

V. THE ANALYSIS OF PERFORMANCE

The convergent graph and data comparison are shown in figure 9 and table 2: The DRRT algorithm is convergent faster than the other three algorithms. After 3000 iterations, the length of its planned path is 264.5750km, the times of searching failure are 41427 times, and the planning time is 1256ms. Compared with the other three algorithms, its performance significantly improved, as well as length of its planned path and its planning time are both shorter than other algorithms. Therefore, DRRT can achieve a better balance between the planning time and the optimal path, and also can meet the real-time requirement for algorithm when the UAV executes task in the complex environment.

FIGURE IX. THE CONVERGENT GRAPH

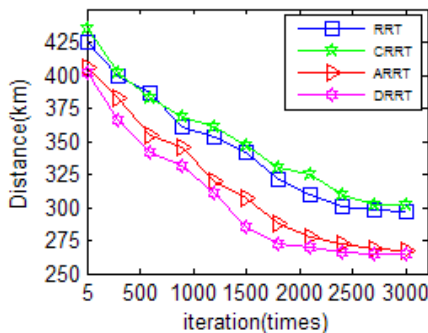


TABLE II. THE COMPARISON OF DATA

Algorithm	Time (ms)	Length (km)	Failed times
RRT	2219	297.3083	76613
CRRT	1172	301.2744	84741
ARRT	1572	267.8498	54042
DRRT	1256	264.5750	41427

VI. CONCLUSION

In this paper, the RRT algorithm for UAVs path planning is discussed. Aiming at these disadvantages which easily falling

into local optimum area and long planning time of RRT algorithm, this paper presents an adaptive RRT algorithm based on dynamic step (DRRT) for path planning and carries on the simulation to prove its effectiveness and feasibility. The simulation result shows that the adaptive RRT algorithm based on dynamic step overcomes these above disadvantages. The distance of DRRT planned path and planning time are obviously less than other algorithms. In addition, the planned path is much closer to the optimal path especially in executing the task in multi-obstacle environment. Because the advantage of RRT is easy to implement in the multi-dimensional environment, the DRRT will be applied into three-dimensional environment in my next work.

REFERENCES

- [1] WEI Ruixuan, LI Xueren. UAV systems and operational use [M]. National Defence Industry Press, 2009, 33-35.
- [2] Junfeng Yao, Chao Lin, Xiaobiao Xie. Path Planning for Virtual Human Motion Using Improved A* Star Algorithm [A]. Information Technology: New Generations [C]. Las Vegas, 2010, 1154 - 1158.
- [3] AVASH RAICHAUDHURI, ASHI JAIN, et al. Genetic Algorithm based Logistics Route Planning [J]. International Journal of Innovation, Management and Technology, Vol. 1, No. 2, June 2010. ISSN: 2010-0248
- [4] RUTGER CLAES, TOM HOLVOET. Ant Colony Optimization applied to Route Planning using Link Travel Time predictions [J]. International Parallel & Distributed Processing Symposium. 2011, 353-359
- [5] Smkuffner JR, J J. Rapidly-exploring random trees: progress and prospects [A]. Proceedings of the Fourth Workshop on the Algorithmic Foundations of Robotics [C]. Natick, 2000, 45-59.
- [6] AMIN J N, BOSKOVIC J D, MEHRA R K. A fast and efficient approach to path planning for unmanned vehicles [R]. AIAA 2006 -6103, 2006.
- [7] Rahul Kala, Kevin Warwick. Multi-vehicle planning using RRT-connect [J]. Paladyn, 2011, Vol 2, Issue 3, 134-144.
- [8] Sertac Karaman Matthew R. Walter, et al. Anytime Motion Planning using the rrt* [A]. The International Journal of Robotics Research [C], 2011, 846-894.
- [9] Hai Zhu Pan, Jin Xue Zhang. Extending RRT for Robot Motion Planning with SLAM. Applied Mechanics and Materials, 151, 493. 2012: 493-497.
- [10] Yoshiaki Kuwata, Justin Teoy, Sertac Karaman, E. et al. Motion planning in complex environments using closed-loop prediction [A]. Guidance, Navigation, and Control Conference [C]. Honolulu, 2008, 7166-7188.
- [11] Luders, B.D., Dept. of Aeronaut., Karaman, S. et al. Bounds on tracking error using closed-loop rapidly exploring random trees [A]. American Control Conference [C]. Baltimore, 2010, 5406-5412.
- [12] B. LUDERS, S. KARAMAN, E. FRAZZOLI, and J. P. How, "Bounds on tracking error using closed-loop rapidly-exploring random trees," in American Control Conference (ACC), Baltimore, MD, 2010, pp. 5406-5412.
- [13] DONG Wenhong, YI Bo, LI Fei. Research on environment model for route planning of UAV [J]. Computer Engineering and Applications, 2012, Vol 15, 236-239.
- [14] Wenping Wang, Helmut Pottmann, Yang Liu. Fitting B-Spline Curves to Point Clouds by Curvature-Based Squared Distance Minimization [J]. ACM Transactions on Graphics, 2006, Vol 25, No2, 214-238.